Roll Number: _____

Thapar Institute of Engineering & Technology, Patiala
Department of Computer Science and Engineering
END SEMESTER EXAMINATION (EST)

BE- 2nd Year COE/CSE
Date: 16th May, 2026
Time: 9:00 AM to 12:00 PM
Max marks: 60 (Weightage: 40)

Course Code: UCS520
Course Name: Computer Networks
Name of Faculty: MCL, SKK, SHD, GVR, HMG, ASB, SUB, TG

Note: Attempt any FOUR questions. Attempt all sub-parts of each question at one place. Mention page no. of your attempt at front page of your answer sheet. Assume missing data (if any).

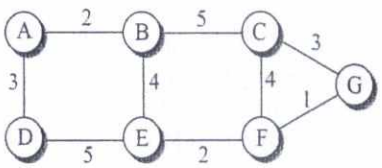
Q.1. **(a)** Given the data word 1101011011 and the divisor 10011, perform Cyclic Redundancy Check (CRC) encoding to generate the transmitted codeword at the sender side. At the receiver side, perform CRC checking using the same divisor and state whether the received frame is error-free or corrupted. (6) CO2 L3

(b) i) A packet arrives at a router and must be forwarded through a network with a Maximum Transmission Unit (MTU) of 576 bytes. The received IP packet contains a 20 bytes header and a 1484 bytes data payload. Assuming that fragmentation is performed at the router, determine the number of fragments generated and complete the following fragmentation table for all fragments. (3+2+4) CO3 L4

	Total Length	Identification No	MF	Fragment Offset
Original Packet	1504	123573	0	0
Fragment1				
…				

ii) An IP datagram is received with the following partial IP header information represented in hexadecimal format: 45000054 00030000 2006...
Analyze the given IP header and determine the following: the size of the IP header, the number of padding bits required if the header size is 22 bytes, the size of the IP payload (data field), and whether the packet is fragmented or not.

iii) Consider the weighted graph shown below Figure a with nodes *A–G* and associated link costs.



a. The weighted graph

Using the Link State Routing approach, with node A as the source and node G as the destination, determine the least-cost path. Show key intermediate cost updates and final path with total cost.

Q.2. **(a)** Five channels, each transmitting at 150 bytes/s, are multiplexed using synchronous TDM. Two bytes from each channel are included in every frame. Calculate the frame size, frame rate, frame duration, and bit rate of the multiplexed link. With the help of suitable diagrams, differentiate between multilevel multiplexing and multiple-slot multiplexing. (4+3) CO2 L3

(b) i) A university campus uses wired Ethernet LANs and wireless Wi-Fi networks. During peak hours, network performance degrades due to collisions and retransmissions. With the help of suitable diagrams, explain the working of CSMA/CD and CSMA/CA. Compare both protocols with respect to their collision handling mechanisms and efficiency under heavy traffic conditions. (5+3) CO2 L3

	ii) In an ALOHA-based communication system, frames of 400 bits are transmitted over a 400 kbps channel. Calculate the throughput for a Slotted ALOHA system with an offered load of 2000 frames/s and for a Pure ALOHA system with an offered load of 500 frames/s.			
Q.3.	**(a)** With the help of suitable diagram, discuss circuit-switched and packet-switched networks. Explain the sequence of logical steps involved in data transmission for both approaches. Also, provide a comparative analysis of these switching techniques in terms of efficiency and delay. **(b)** A network engineer proposes a design in which the application layer performs compression, encryption, routing, and delivery. Data Link and Network layers are removed to reduce overhead. The design also relies solely on physical addressing for communication across networks and assumes that the TCP/IP and OSI models are structurally identical. i) Identify three major conceptual errors in this design and explain each clearly. ii) Explain why layer separation is essential for scalability and interoperability in networks. iii) Explain why physical addressing alone cannot support communication across multiple networks. iv) Give one practical difference between TCP/IP and OSI that would directly affect this design.	(8) (3+2+1+1)	CO1 CO1	L4 L4
Q.4.	**(a)** Draw and explain the format of a TCP segment header and SCTP frame header along with their fields in detail. Discuss the significance of sustainability in congestion control mechanisms in modern communication networks and list the sustainable congestion control techniques used in such networks. **(b)** i) In a distributed network environment, Client 1 sends an e-mail to Client 2. Identify the application layer protocols used for e-mail transmission and retrieval. Explain the functionality of each protocol involved in the communication process. ii) Identify and discuss at least four key challenges associated with the design of sustainable and energy-efficient distributed networking applications. Also, suggest suitable solutions to address each identified challenge.	(5+2) (4+4)	CO1 CO2	L2 L2
Q.5.	**(a)** An ISP assigns the IPv4 block 142.150.235.0/27 to a research lab, with default gateway 142.150.235.1 and broadcast address 142.150.235.31. Answer the following: i) What is the assigned subnetwork and extended prefix? ii) What usable IP addresses are available for the lab computers? ii) What IP, default gateway, and subnet mask should be configured on a computer? iv) How many host bits identify the subnet? v) What is the role of the default gateway? Why is this information important? vi) Is a router required in the lab? Justify your answer. vi) Split the block into 3 subnetworks, one with 16 addresses and two equal-sized subnetworks, and give each network address, usable range, and host count. **(b)** Consider a network with five nodes (N1–N5) operating under the Distance Vector Routing protocol. The stabilized distance vectors are as follow: N1(0, 1, 7, 8, 4); N2(1, 0, 6, 7, 3); N3(7, 6, 0, 2, 6); N4(8, 7, 2, 0, 4); N5(4, 3, 6, 4, 0). Assuming symmetric link costs, determine: i) If the cost of link N2-N3 decreases to 2, what is the new vector at N3 after one round? ii) If link N1–N2 goes down and N2 sets that entry to ∞ what is the cost to N1 in N3's vector after the next round?	(6+4) (2.5+2.5)	CO3 CO3	L4 L3

********** All the Best **********